



GROW_SCI_A2_W05 · GROW Science teacher note

Staff-facing; review only. Verbatim SoW objective: Consolidate forces, space and climate; answer scientific questions.

Source: GROW scheme of work, Autumn 2 week 5. Pupil goal: choose a model that fits a science question and use a matching clue.

40-minute sequence and success

Arrival 5 · Starter 3 · I do 5 · We do 7 · We do 2 5 · Independent 8 · Review 4 · Exit 3 = 40.

One-minute resets are inside We do 2, Review and Exit. Success: I can match a question to a model., I can give a claim with a matching clue., I can say one thing the model cannot show.

Retrieval and I-do check

When	Question	Answer	SoW cell
last lesson	Give one link in the warming chain.	Burning fossil fuel adds carbon dioxide; extra greenhouse gas slows heat loss.	GROW scheme of work, Autumn 2 week 4
2-4 weeks	Name one fact our Solar System cards support.	The Sun is a star; Earth orbits it; or the Moon orbits Earth.	GROW scheme of work, Autumn 2 week 3
last term	How can friction affect a moving toy?	It can slow it down.	GROW scheme of work, Autumn 1 week 3

Autumn 1 is the prior half-term. Confirm actual taught coverage before using a retrieval question. Show me: If my question is about day and night, should I point to the toy or torch-and-ball? Answer: Torch and ball. All pupils show by point, card or board.

Words (same as page and organiser):

Word	Tier	Simple definition	Example sentence
force	Tier 3	a push or pull on an object	Friction is a force that can slow a toy.
rotation	Tier 3	a turn around an axis	Earth's rotation brings day and night.
evidence	Tier 2	an observation or result that supports a claim	The shorter run is evidence for my answer.
climate	Tier 3	weather patterns over many years	A hot day alone does not show a climate trend.

Faded example, tiered task and stretch

I do full: "Toy motion question → choose toy and surface → measure distance → test friction as one cause." We do part-done: "Day and night question → choose torch and ball → observe → explain Earth's turn." Removed step: Name the observation of the marked place moving away from the light. Independent: no filled step. Next fade: Next time omit the model choice, so the pupil chooses a suitable model before looking for a clue.

We do F: Show the torch and marked ball; point to the place that moves into shade. M: Give an answer and a matching clue. H: Explain why a toy run cannot test a climate claim even when it gives



a clear result. Independent F: Match a question to a model. M: Give an answer and a matching clue. H: Compare two models, clues and limits. Go deeper: Compare two models: what can each show and miss? Named harder stretch — Model limits: Two toy runs differ on one surface. Can friction alone explain this? Choose one control that would help.

The model, source or practical route is: toy and two surfaces, torch and ball, warming cards. Two on-screen interactions are card sort and What helps test a toy-motion claim?; native keyboard controls, live feedback, no storage. The diagram is original and simplified, not to scale. Its labels are checked in the facts ledger.

Practical: Predict first: Which surface may let the same toy travel farther? Kit: Small toy, Rough and smooth paper, Start line, Ruler, Torch and ball, Warming cards. Steps: Mark one start and use the same toy; Launch twice on each surface in the same way; Measure each run and record it; Use other models only for their own questions. Results table: Surface | run 1 | run 2, Rough | _ | _, Smooth | _ | _. Check against the school risk assessment and CLEAPSS guidance. Keep kit at desk height and torch light out of eyes. Do not infer a cause from one uncontrolled run.

Hinge and decision

Question: A model trolley stops sooner on a mat than on a smooth board. Which idea is relevant to test?

Option	Correct?	Diagnosis
The length of Earth's orbit.	no	Transfers a space model to a surface-and-motion question.
Friction at the trolley surface.	yes	Uses the new case accurately.
The amount of daylight outside.	no	Uses daylight rather than the changed surface to explain the trolley run.
The climate of the whole Earth.	no	Uses a whole-Earth climate pattern for a local trolley test.

Rule: About 3 in 4 or more right → move on. Otherwise re-model with the named I-do example, then ask the parallel hinge. Parallel: A block is pulled over two surfaces. Which idea helps test why the pull differs: friction or Earth's rotation? Answer: friction.

Misconceptions:

Misconception	Spot	Correct
Any science model can answer any question.	Pupil chooses A, a space model for trolley motion.	Match the model's observable feature to the question before making a claim.
One different toy run proves roughness alone caused it.	Pupil chooses B correctly but then claims one run proves the cause without controls.	Repeat runs with a consistent start and push before treating the difference as strong evidence.

AQA Entry Level Science style bridge (classroom planning): Choose a suitable model and describe the observation that supports a claim. This is a planning bridge, not a claimed unit outcome or code.

Feedback codes: VF · WS · I · NS(+) · E · R · // · ?. At We do 2, record WS or I and actual prompt level; at Review, VF and specific NS(+); at Exit, one self-edit // or read back ?. R means a response to the next step heard by an adult in the next lesson, never auto-awarded. E: evidence captured on the school's evidence app only if an observable curriculum action occurred, never while distressed. The page stores no response. Pathways: high support with VF/WS; Build: mix of WS/I; Grow: mostly I and self-edit; Launch: independent self-assessment.

Planned adaptations



Broad area	EEF five-a-day strategy	Specific support
Cognition & Learning	scaffolding	Use three labelled kit photos matched to three question cards, then remove labels.
Communication & Interaction	cognitive/metacognitive strategies	Let a partner rehearse claim and clue with the M/H frame before recording.
SEMH	flexible grouping	Offer a known station role and a quiet check-in before changing stations.
Sensory/Physical	technology	Keep kits on a fixed tray and allow a staff-captured photo of a real demonstration.
EAL (language, not SEN)	explicit instruction	Pre-teach force, rotation and climate with icons; allow home-language planning and M/H explanation frames.

Keep M/H challenge open for EAL with the same words and home-language planning. Pupil may point, sign, speak, draw, type, write, use a scribe or have a staff-captured photo. Offer speech-to-text, TTS or immersive reader on the staff device.

Adults, regulation and prompt ladder

Wait and count to 5. → Point to the lesson's diagram, kit or source. → Ask: What is done, and what is the next step? → Model one step, then return the next step to the pupil. → Finish one example together, then fade the help. No physical prompt. Stage scripts: I do "Listen: first I read the question, then choose the matching kit."; We do "Which model is already chosen? What observation is missing?"; Independent "Show me your claim; which clue did you find on your own?"

At Independent, one adult works with supported pupils, the other holds the group; choose roles at the start and swap across the week. If only one adult attends, use the actual staffing plan. At Exit, hand off the deepest prompt level and what was independent.

Cold-call / invite bank, answers:

- F: Which kit is for a motion question? → Toy and surface.; Which kit is for day and night? → Torch and ball.
- M: Why do we name a clue? → It supports the claim.; Why is one toy run weak evidence? → The start or push might have changed.
- H: What can the torch-and-ball model not show exactly? → True size and distance, among other details.; Why does a warming card not show a local force? → It represents a long-term causal chain, not the toy's motion.

Regulation: Agree a quiet pause signal and supervised place. Check in before hinge and Independent; point to the last completed step on re-entry. No evidence while a pupil is distressed.

lesson record: GROW_SCI_A2_W05 via staff device or projector only. Gatsby 4: Engineering technician: choose a test and match each claim to a result before changing a design.

Reading gate: GROW 8–11; rough visible-body ARI grade 3.33; age guide 9.33 years; rough FK grade 3.48; 822 pupil-visible words excluding marked adult guidance and navigation. Screening estimate only. This is a rough screen, not an individual assessment.

Pupil-facing risk: A moving toy may hit someone or fall from the table.

Pupil-facing safety: Use a low clear track. Keep hands and faces away from the run.



SLT planning and evaluation

Lesson code: SCI-GROW-A2-W05

SoW reference: GROW scheme of work, Autumn 2 week 5

Exact SoW outcome: Consolidate forces, space and climate; answer scientific questions.

Evidence (scheme of work): Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Evidence in this lesson: Pearson Edexcel Entry Level Certificate in Science 8939 (E1→E3); AQA UAS science units. F: working-scientifically observation & results tables. S: end-of-unit assessment + Entry Level / UAS evidence.

Where it shows in books: Independent observation/result table or labelled explanation; review edits and exit response beside the original work.

Label evidence on the school's evidence app with this code: SCI-GROW-A2-W05. For spoken, practical or photographic evidence, record the same code with the evidence.

Arrival (5 min): Match each question to a kit card: toy and surface, globe and torch, or warming chain. Staff: Offer a predictable seat or standing spot and the first visual cue. No public test.

Starter (3 min): Think back. Say, point or write one answer for each question. Staff: Ask three low-stakes retrieval questions; prior coverage must be checked.

I Do (5 min): I choose a toy for a motion question. The toy went less far on rough paper in this run. I must repeat before I claim why. Staff: Model the full reasoning step, pre-teach four words and ask the show-me question.

We Do (7 min): Torch and ball are chosen for day and night. Add the missing clue. Say why it fits. Staff: Show the part-done example, then allow paired talk before an answer. One-minute reset inside this stage.

We Do 2 (5 min): A toy travels less far on rough paper in this example. Choose the best idea to test. Staff: All pupils answer the four-option hinge; use the decision rule, not a score.

Independent (8 min): Choose two stations. Give a claim and matching clue for each. Name one limit. Staff: One adult supports agreed pupils, the other holds the group; swap across the week. One-minute reset inside this stage.

Review (4 min): Does your clue fit? Fix one clue or add a limit. Staff: Give a specific verbal comment (VF) and invite one self-edit (//) or read-back (?).

Exit (3 min): Name a model, claim and clue. Say what changed after feedback. Staff: Hear the answer and name how it changes the next example or support. One-minute reset and handoff.

Planned adaptation — cognition learning: Use three labelled kit photos matched to three question cards, then remove labels.

Planned adaptation — communication interaction: Let a partner rehearse claim and clue with the M/H frame before recording.

Planned adaptation — semh: Offer a known station role and a quiet check-in before changing stations.

Planned adaptation — sensory physical: Keep kits on a fixed tray and allow a staff-captured photo of a real demonstration.

Planned adaptation — eal: Pre-teach force, rotation and climate with icons; allow home-language planning and M/H explanation frames.



Planned adaptation — eef strategy: cognition learning: scaffolding; communication interaction: cognitive/metacognitive strategies; semh: flexible grouping; sensory physical: technology; eal: explicit instruction

Staff: confirm the planned adaptations actually used and record any changes below.

If behind — must teach core: Choose two stations. Give a claim and matching clue for each. Name one limit. Preserve the individual evidence required above.

If behind — shorten or merge: After a secure hinge, shorten repeated oral rehearsal. If needed, merge the exit reflection into the Arrival of Autumn 2 W06. Keep the core evidence and any safety teaching; record the change and coverage impact.

If behind — next lesson: Autumn 2 W06: Consolidate and revisit key ideas through investigation.

If behind — dependency: Pass on today's checked understanding and unresolved hinge errors for the next starter. Do not replace incomplete individual work with a group product.

Staff evaluation — date taught:

Staff evaluation — delivered as planned: ☐ Yes ☐ Partly ☐ No

Staff evaluation — criteria met: ☐ All ☐ Most ☐ Some

Staff evaluation — adaptations made and why:

Staff evaluation — change from sow reason and coverage impact:

Staff evaluation — evidence banked on the school's evidence app:

Individual target opportunities

Cognition & learning: We do: Name the observation of the marked place moving away from the light. Record the pupil's own sequencing, attention or independence target, including the prompt level needed.

Communication & interaction: Check: respond to "A model trolley stops sooner on a mat than on a smooth board. Which idea is relevant to test?" in the pupil's agreed mode. Evidence an individual communication/turn-taking target without requiring speech.

SEMH: Arrival: Match each question to a kit card: toy and surface, globe and torch, or warming chain. Use the agreed predictable start or pause/re-entry plan; evidence the pupil's own regulation or engagement target privately.

Sensory/physical: Independent: Choose two stations. Give a claim and matching clue for each. Name one limit. Offer the agreed accessible input, enlarged material or adapted equipment; record the pupil's target-related access/independence, not a diagnosis.

EAL: Model/shared rehearsal: use force, rotation, evidence, climate with the actual example; allow first-language planning and oral rehearsal before the independent response. Record the agreed language target; EAL is not SEN.

Capture E against the pupil's own IEDP target on the school's evidence app; the IEDP carries their EHCP-linked and Boxall targets, so one record serves every plan. Follow the pupil's IBSP where it differs from these routines.